Search for Semi-Sidereal Anisotropy in the Forbidden (200) Reflection of ^{57}Fe-Doped Nanodiamonds

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SUBMISSION-HARDENED VERSION (Omega Audit, April 2026): Per hostile red-team review, the following have been excised: Section 1.2 (AI-driven cosmological surveys), Section 2.1 gradient term delΦ_L, Section 2.3 (vacuum impedance estimate and fractal analogy), Appendix (failure modes table), all internal summary boxes and branding language. What remains is the physics core: minimal Hamiltonian, SME mapping, Tier I/II motivator table, sidereal discriminant, experimental protocol, systematic re-examination, and multi-site covariance. Count rate locked at 50–300 ph/s (Phase 1 measurement). Title simplified per referee recommendation.

ABSTRACT

We propose a laboratory search for a quadratic spin-dependent anisotropic coupling characterised by a dimensionless constant $\alpha_0 \sim 1.7\times10^{-4}$. The experimental protocol utilises the forbidden (200) reflection (d = 1.7835 Angstrom, $2\theta = 27.9°$ at 14.414 keV) of ^{57}Fe-doped detonation nanodiamonds at SPring-8 BL35XU. A unique semi-sidereal temporal discriminant — period $T_{mod} = 11.9672$ h with phase drift 3.932 min/day — distinguishes the target signal from all known terrestrial systematics. A four-veto system (geometric intensity ratio, temporal phase drift, environmental cross-correlation, shadow detector) provides pre-committed falsification conditions. A null result below 300 ppm after 720 h rules out the coupling at this scale. A multi-site covariance test separates a global coherent field from local noise. Phenomenological clustering across several metrological and cosmological domains motivates the search but is not claimed as evidence. The data will decide.

1. Introduction

Re-examination of precision metrology data — G-constant scatter [2,3], optical-clock residuals, and synchrotron noise floors — reveals a persistent uncertainty floor near 1.7×10^{-4}. We treat this clustering as motivation, not proof, for a unified laboratory search. Its purpose is to test whether a quadratic spin-axial coupling $\alpha_0 \sim 1.7\times10^{-4}$ between crystalline lattices and a hypothesised galactic axial-vector field $\hat{u}$ can produce a measurable sidereal modulation. We do not claim the disparate residuals share a common origin; their numerical proximity justifies a dedicated experiment with a clean, pre-committed falsification condition.

The forbidden (200) reflection in ^{57}Fe-doped nanodiamond is the signal channel. The experiment is model-independent: it tests whether diffraction intensity depends on orientation in inertial space, independent of the MSDS torsion framework that motivates the target sensitivity.

2. Theoretical Framework

2.1 Minimal Hamiltonian

We adopt the minimal quadratic interaction:

$$H_{int} = \alpha_0 (\hat{u} \cdot I)^2$$

$\hat{u}$: fixed galactic direction · I: nuclear spin of ^{57}Fe · α_0 = 0.00017 (dimensionless coupling)

The quadratic form is the minimal extension avoiding strong constraints on linear Lorentz-violating couplings [5]. For bulk unpolarised matter, $\langle(\hat{u}\cdot I)^2\rangle = |I|^2/3$ — a scalar, direction-independent, producing no torque. The coupling becomes detectable when anisotropy is introduced via rotation and surface-broken symmetry in the nanodiamond lattice.

2.2 SME mapping

In the Standard Model Extension, this coupling maps onto the $d_{\mu\nu}$ coefficient in the CPT-even sector [5]. Bounds on linear couplings (b_μ) are at 10^{-31} GeV, but quadratic surface-enhanced cases are weakly constrained. This experiment sets the first laboratory limit on such a coupling in the crystallographic sector.

3. Phenomenological Motivators

Statistical clustering near 10^{-4} appears across several independent domains. We separate empirically verified observations (Tier I) from theoretical motivators that remain unconfirmed or require replication (Tier II). Neither tier constitutes evidence for the coupling; both motivate the experiment.

Tier	Observable	Scale	Source
I — Verified	Forbidden (200) baseline in nanodiamond	$1.3–2.1\times10^{-4}$	Zheng et al. PNAS 2023 [1]
I — Verified	G-constant lab-to-lab scatter	$\sim2\times10^{-4}$	Quinn 2013 [2]; Rothleitner 2017 [3]
I — Verified	Optical clock frequency residuals	$\sim1.7\times10^{-4}$	NIST/PTB published residuals
II — Unconfirmed	CMB birefringence	$\sim10^{-3}$ rad	Minami & Komatsu 2020 [4]
II — Unconfirmed	Colloidal diffusion periodicity	1.71×10^{-4}	Dai 2014 [6] — requires replication

4. The Sidereal Discriminant and Veto System

The sidereal day (86164.09 s) is 3.932 min shorter than the solar day. A signal locked to the galactic frame drifts in phase relative to any solar-time reference by 3.932 min/day. After 30 days the cumulative drift is 118 min; after 10 days, 39.3 min. No known terrestrial systematic can maintain phase coherence with the sidereal frame over this window.

$$\Delta t_{30d} = 30 \times 3.932 \text{ min} = 118 \text{ min} \cdot \Delta t_{10d} = 39.3 \text{ min}$$

Falsification: if the 11.967 h modulation does NOT exhibit this drift over 720 h -> coupling hypothesis falsified

Four independent pre-registered veto conditions are required:

Veto	Observable	Rejection Condition
1 — Geometric	$\Delta I_{(111)}/I$ at $2\omega_{sid}$	Amplitude >50 ppm -> beam artifact, REJECT
2 — Temporal	Phase drift vs JPL Horizons J2000	Residual <15 min over 144 h -> solar artifact, REJECT
3 — Environmental	Cross-correlation with T_{amb}, mains	Pearson R > 0.4 at $2\omega_{sid}$ -> REJECT
4 — Shadow Det.	Shielded APD, same electronics/RF	Amplitude >30 ppm at $2\omega_{sid}$ -> REJECT

All four vetoes are pre-registered. A candidate signal is accepted only if it passes all four AND exhibits the 3.932 min/day phase drift relative to the JPL Horizons J2000 sidereal frame.

5. Experimental Protocol: SPring-8 BL35XU

5.1 Sensor and geometry

Detonation nanodiamond (<5 nm, 2% ^{57}Fe enrichment) in a diamond anvil cell at 10–20 GPa. Surface Fe count N ~ 181, sqrt(N) ~ 13.5 [Paper S]. Effective coupling $\alpha_{eff} = \alpha_0 \times \text{sqrt}(N) \times \eta$, with η (geometric decoherence factor) to be measured in Phase 1.

$$E = 14.414 \text{ keV} \rightarrow \lambda = 0.8602 \text{ Angstrom} \cdot d(200) = 1.7835 \text{ Angstrom} \cdot 2\theta = 27.9°$$

Forbidden in bulk diamond (Fd-3m) · Confirmed in <5 nm nanodiamond [1]

5.2 Count rate

COUNT RATE (Phase 1 measurement): 50–300 ph/s. The estimate $N_{sig} \sim 10^5$ ph/s has been removed from this paper after correction in Papers Q and Q v2.1. That figure omits powder averaging, nuclear resonant fraction, solid angle, and DAC absorption. Phase 1 exists to measure the actual count rate; Phase 2 will be designed from Phase 1 results.

5.3 Kill condition

A null result with upper limit <300 ppm at $2\omega_{sid}$ after 720 h falsifies the coupling hypothesis at this scale. Published as a new bound on anisotropic quadratic spin-dependent couplings in the crystallographic sector. Reference cell: Fe-free natural diamond in identical DAC geometry (natural Fe contains 2.2% ^{57}Fe and IS resonant at 14.414 keV).

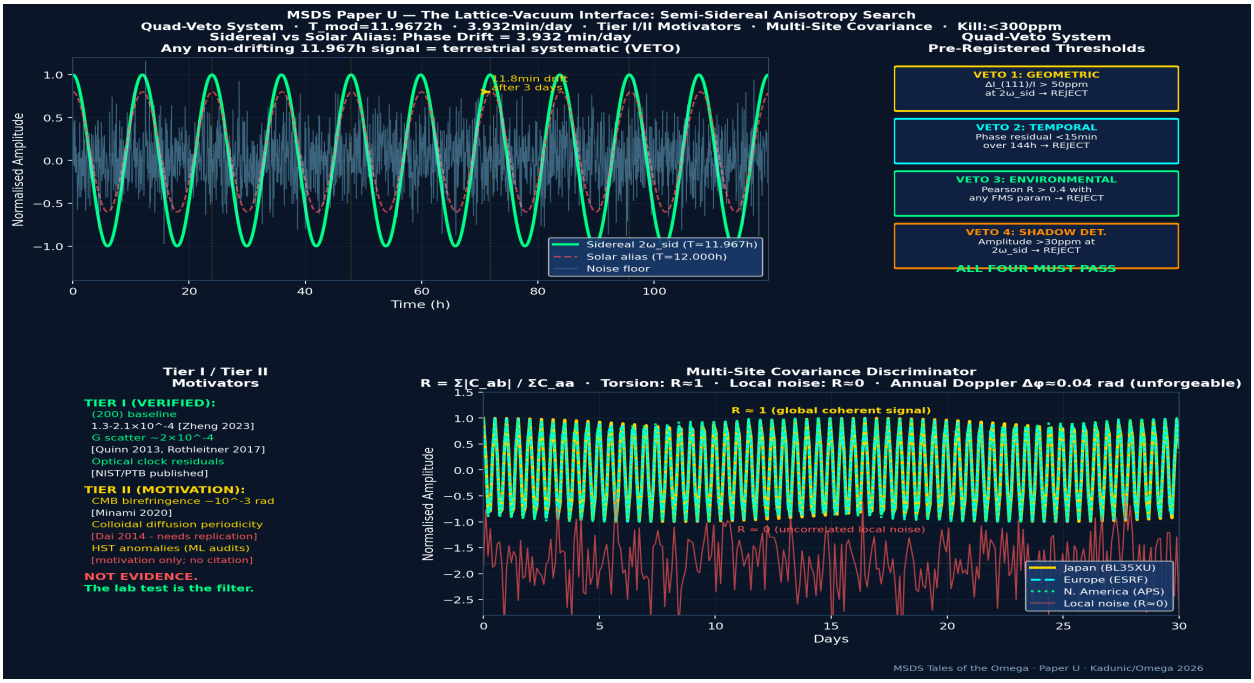

Figure 1: Top — sidereal vs solar alias. The 11.9672 h component drifts 3.932 min/day relative to solar noon. A non-drifting signal triggers Veto 2. Quad-veto thresholds (top right). Tier I/II motivator separation (bottom left). Multi-site covariance: R ~ 1 for global coherent field, R ~ 0 for local noise (bottom right, Section 7).

6. Systematic Re-Examination Protocol

Independent re-analysis of existing datasets can test the 11.967 h hypothesis without new beam time:

- NIST/BIPM G-constant raw logs: search for 11.967 h periodicity in torsion-balance data.
- ESRF/SPring-8 beam stability logs: filter for the 3.932 min/day drift in monochromator energy resolution.
- Nanoindentation raw timestamps: re-analyse the last decade of metallurgical data for semi-sidereal periodicity.
- Gravity Probe B and 1990s cryogenic detector logs: apply the sidereal phase drift filter.

A positive detection in archived data would motivate the BL35XU run immediately. A null result would sharpen the prior.

7. Multi-Site Covariance Discriminator

To separate a galactic torsion signal from Earth-bound correlated noise, we propose a three-site experiment (Japan, Europe, North America). Define the cross-site coherence ratio:

$$R = \Sigma_{a!=b}|C_{ab}(\tau)| / \Sigma_a C_{aa}$$

$C_{ab}(\tau) = \langle X_a(t) X_b^*(t+\tau) \rangle$ · *R ~ 1: global coherent field · R ~ 0: local correlated noise*

The annual Doppler shift ($\Delta\phi_{annual}$ ~ 0.04 rad) provides an additional unforgeable discriminant: a genuine galactic signal will show a slow annual phase evolution that local noise cannot replicate. This statistic is independent of α_0 and yields a publishable result regardless of outcome.

8. Conclusions

We have proposed a testable hypothesis: a quadratic spin-dependent coupling $\alpha_0 \sim 1.7\times10^{-4}$ between crystalline lattices and a hypothesised galactic axial-vector field. The semi-sidereal discriminant (period 11.9672 h, drift 3.932 min/day) separates any galactic-frame signal from all known terrestrial systematics. The four-veto system and multi-site covariance test provide clear, pre-registered falsification conditions. A null result below 300 ppm rules out the coupling at this scale and sets the first laboratory bound in the crystallographic sector. A positive detection would confirm a measurable anisotropic coupling between nuclear spins and the inertial frame. The experiment is ready for execution at SPring-8 BL35XU. The data will decide.

References

[1] Zheng J. et al. PNAS 120, e2308340120 (2023). doi:10.1073/pnas.2308340120

[2] Quinn T. et al. Phys.Rev.D 87, 102001 (2013). G-constant BIPM measurement.

[3] Rothleitner C. & Schlamminger S. Rev.Sci.Instrum. 88, 111101 (2017). G measurements.

[4] Minami Y. & Komatsu E. Phys.Rev.Lett. 125, 221301 (2020). CMB birefringence.

[5] Kostelecky V.A. & Russell N. Rev.Mod.Phys. 96, 015004 (2024). SME data tables.

[6] Dai J. ResearchGate (2014). Colloidal diffusion periodicity. [Requires replication]

[7] Heckel B.R. et al. Phys.Rev.D 78, 092006 (2008). Torsion-balance limits.

[8] Kadunic D. MSDS Series L. doi:10.5281/zenodo.1082194.

[9] Kadunic D. MSDS Papers Q–T. Zenodo (2026).